Medicare Part A
Pre-Payment Edit Simulation

Implementation Plan - POC

Snowflake | Near Real-Time | SNIP-Aligned DQ | 11 Edit Rules

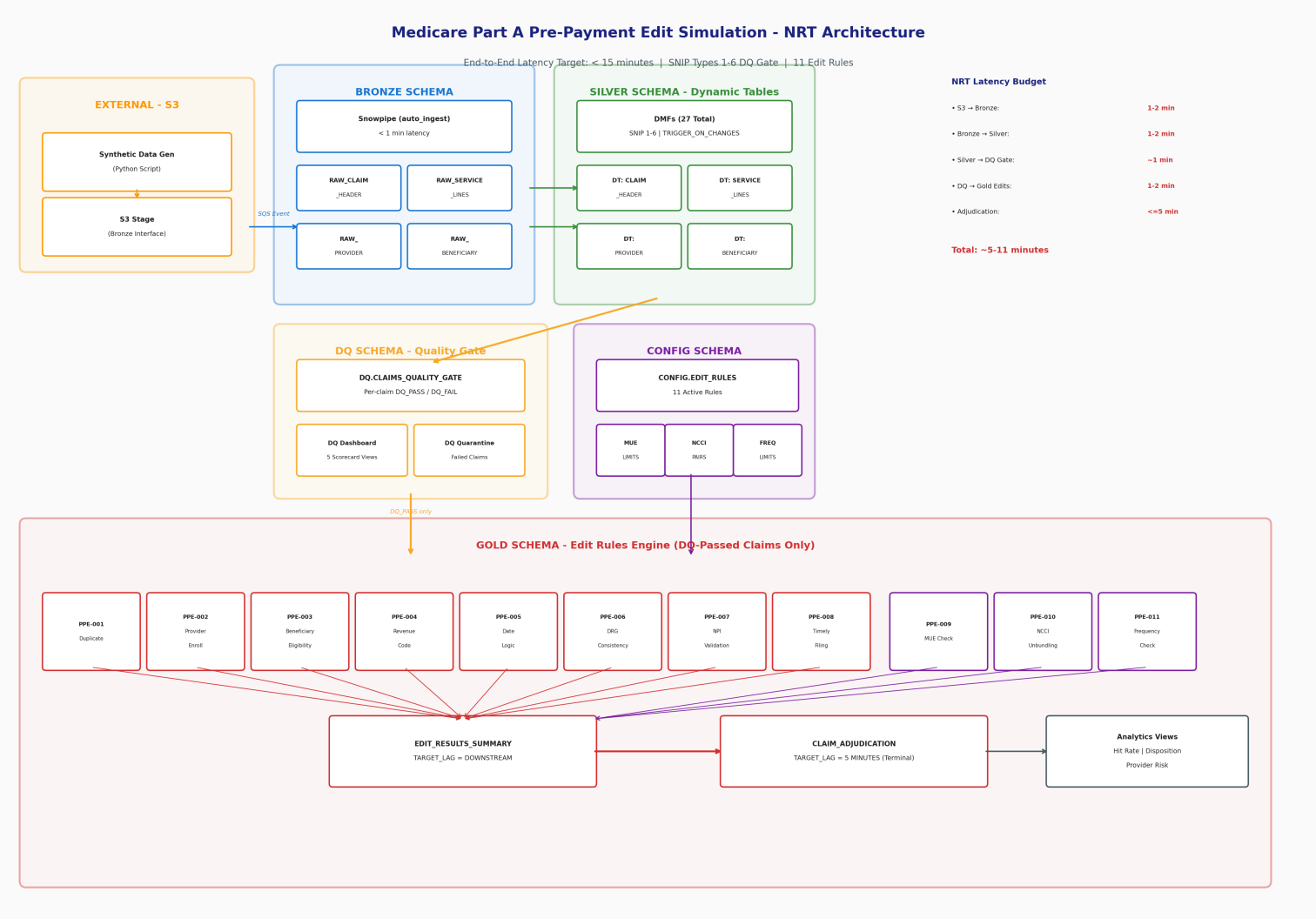
Medallion Architecture (Bronze / Silver / Gold)

Date: March 2026

Platform: Snowflake (Dynamic Tables, Snowpipe, DMFs)

1. Architecture Overview

The architecture follows a Medallion pattern (Bronze/Silver/Gold) with near real-time processing. Data flows from S3 through Snowpipe into Bronze, transforms via Dynamic Tables into Silver, passes through a SNIP-aligned DQ gate, and is adjudicated by 11 edit rules in the Gold layer.



NRT Latency Budget (Target: < 15 minutes)

Stage	Mechanism	Expected Latency
S3 -> Bronze	Snowpipe (SQS auto_ingest)	1-2 minutes
Bronze -> Silver	Dynamic Tables (DOWNSTREAM)	~1-2 minutes
Silver -> DQ Gate	Dynamic Table (DOWNSTREAM)	~1 minute
DQ -> Gold Edit Rules	Dynamic Tables (DOWNSTREAM)	~1-2 minutes
Adjudication (Terminal)	Dynamic Table (5 MIN lag)	<= 5 minutes
TOTAL		~5-11 minutes

2. Key Design Decisions

Snowpipe (NRT Ingestion)

Snowpipe with AUTO_INGEST=TRUE and SQS event notifications replaces scheduled COPY INTO Tasks. Sub-minute ingestion latency.

TARGET_LAG = DOWNSTREAM

All Silver and intermediate Gold Dynamic Tables use DOWNSTREAM lag. A single terminal DT (CLAIM_ADJUDICATION, 5 min) drives the entire cascade.

All-SQL Edit Rules

All 11 edit rules are expressed as pure SQL Dynamic Tables. No Snowpark stored procedure in the hot path.

SNIP 1-6 as DMFs in Silver

27 Data Metric Functions aligned to the SNIP framework validate data quality before claims enter the Gold edit rules engine.

Payer-Specific Edits in Gold

MUE, NCCI unbundling, and frequency limit checks are business rules that require reference table lookups. They live in the Gold layer as edit rule DTs.

DQ Gate (Quality Filter)

A Dynamic Table (DQ.CLAIMS_QUALITY_GATE) evaluates per-claim SNIP checks. Only DQ_PASS claims flow to edit rules. DQ_FAIL claims get DQ_REJECT disposition.

3. Task 1: Snowflake Infrastructure

Database & Schemas

Schema	Purpose
BRONZE	Raw ingestion (VARIANT + metadata)
SILVER	Cleansed/conformed Dynamic Tables
GOLD	Business rules, adjudication, analytics
DQ	Data quality artifacts, DQ gate, dashboard views
CONFIG	Reference/config tables (edit rules, MUE, NCCI, freq limits)

Warehouses

Warehouse	Size	Purpose
CLAIMS_INGEST_WH	X-Small	Manual catch-up only (Snowpipe uses own compute)
CLAIMS_TRANSFORM_WH	Small	Dynamic Tables (Silver + Gold), Snowpark
CLAIMS_ANALYTICS_WH	Small	Analyst queries on Gold views, DQ dashboard

RBAC Roles

Role	Permissions
CLAIMS_ADMIN	Full ownership of all POC objects
CLAIMS_ENGINEER	Read/write on BRONZE/SILVER, execute on GOLD procedures
CLAIMS_ANALYST	Read-only on SILVER/GOLD/DQ (includes DQ dashboard)

4. Task 2: Synthetic Data Generation

Generate realistic synthetic Medicare Part A institutional claims data written as JSON to S3. ~15% of claims are intentionally seeded with errors spanning all SNIP categories and payer edits.

Claim Header Fields

Field	Description	SNIP Relevance
CLAIM_ID	Unique claim identifier	-
DCN	Document Control Number	-
BENEFICIARY_HIC	Medicare Beneficiary Identifier (MBI)	SNIP 1-2: MBI format
PROVIDER_NPI	Billing provider NPI (10-digit)	SNIP 1-2: NPI format
ADMISSION_DATE	Date of admission	SNIP 1-2: Date format
DISCHARGE_DATE	Date of discharge	SNIP 3-4: Date consistency
DRG_CODE	Diagnosis Related Group	SNIP 5-6: Code validity
PRINCIPAL_DIAGNOSIS_CODE	ICD-10-CM primary	SNIP 5-6: ICD-10 format
TOTAL_CHARGES	Total billed amount	SNIP 3-4: Financial balance
TYPE_OF_BILL	3-digit type of bill code	SNIP 1-2: Format
ADMISSION_SOURCE_CODE	Admission source (NEW)	SNIP 3-4: Conditional field
PATIENT_SEX	M/F/U (NEW)	SNIP 5-6: MCE age/sex

Data Volumes

Entity	Volume	Notes
Claim Headers	~50,000	~15% seeded errors
Service Lines	~200,000	~4 lines per claim avg
Providers	500	Mix of Active/Inactive/Revoked
Beneficiaries	30,000	Mix of Active/Terminated
MUE Limits	~200	HCPCS -> max units per day
NCCI Code Pairs	~500	Bundled code pairs
Frequency Limits	~100	HCPCS -> max per period

5. Task 4-5: SNIP-Aligned DMFs (27 Total)

All DMFs fire on TRIGGER_ON_CHANGES schedule. Results are stored in SNOWFLAKE.LOCAL.DATA_QUALITY_MONITORING_RESULTS and power the DQ dashboard.

SNIP Type 1-2: EDI & Syntax Validation (12 DMFs)

#	DMF	Column(s)	What It Measures
1	NULL_COUNT (built-in)	claim_id	NULL primary keys
2	NULL_COUNT (built-in)	provider_npi	Missing NPI
3	NULL_COUNT (built-in)	beneficiary_hic	Missing MBI
4	NULL_COUNT (built-in)	admission_date	Missing admission date
5	DUPLICATE_COUNT (built-in)	claim_id	Duplicate claim IDs
6	BLANK_COUNT (built-in)	drg_code	Empty DRG codes
7	FRESHNESS (built-in)	_LOAD_TIMESTAMP	Data staleness
8	ROW_COUNT (built-in)	claim_id	Volume anomaly detection
9	CHECK_NPI_FORMAT (custom)	provider_npi	NPIs not 10 digits
10	CHECK_MBI_FORMAT (custom)	beneficiary_hic	MBIs not 11-char CMS pattern
11	CHECK_TYPE_OF_BILL_FORMAT	type_of_bill	TOBs not 3-digit numeric
12	CHECK_DATE_FORMAT_VALID	admission/discharge	Future-dated beyond submission

SNIP Type 3-4: Balancing & Situational Logic (4 DMFs)

#	DMF	Column(s)	What It Measures
13	CHECK_DATE_CONSISTENCY	admission/discharge	Discharge before admission
14	CHECK_TOTAL_CHARGES_RANGE	total_charges	Charges <= 0 or > $10M
15	CHECK_FINANCIAL_BALANCE *	CLAIM_HEADER + SERVICE_LINES	SUM(line_charges) != total_charges
16	CHECK_INPATIENT_ADM_SRC	type_of_bill, adm_source	Inpatient TOB (11x) missing admission source

* Multi-table DMF using Snowflake cross-table argument feature

SNIP Type 5-6: Code Set & Clinical Integrity (4 DMFs)

#	DMF	Column(s)	What It Measures
17	CHECK_ICD10_FORMAT	principal_diagnosis_code	ICD-10-CM not valid format (alpha start, 3-7 chars)
18	CHECK_REVENUE_CODE_FORMAT	revenue_code (SVC_LINES)	Revenue codes not 4-digit numeric
19	CHECK_HCPCS_FORMAT	hcpcs_code (SVC_LINES)	HCPCS not valid 5-char format
20	CHECK_AGE_SEX_CONSISTENCY *	CLAIM_HDR + BENEFICIARY	Maternity DRG w/o female; Peds DRG w/ age>=18

* Multi-table DMF using Snowflake cross-table argument feature

Referential Integrity & Completeness (7 DMFs)

#	DMF	Table	What It Measures
21	NULL_COUNT (built-in)	SERVICE_LINES.claim_id	Orphaned service lines (null FK)
22	NULL_COUNT (built-in)	SERVICE_LINES.revenue_code	Missing revenue code
23	CHECK_REF_INTEGRITY *	SVC_LINES -> CLAIM_HDR	Service lines with no matching claim
24	NULL_COUNT (built-in)	PROVIDER.provider_npi	NULL provider PK
25	DUPLICATE_COUNT (built-in)	PROVIDER.provider_npi	Duplicate provider records
26	NULL_COUNT (built-in)	BENEFICIARY.beneficiary_hic	NULL beneficiary PK
27	DUPLICATE_COUNT (built-in)	BENEFICIARY.beneficiary_hic	Duplicate beneficiary records

DMF Summary by SNIP Category

SNIP Category	Built-in	Custom	Total

SNIP Category	Built-in	Custom	Total
SNIP 1-2: EDI & Syntax	8	4	12
SNIP 3-4: Balancing & Situational	0	4	4
SNIP 5-6: Code Set & Clinical	0	4	4
Referential Integrity	5	1 (+ 1 pipeline)	7
TOTAL	14 (incl 1 FRESHNESS)	13	27

6. DQ Gate & Dashboard

DQ Quality Gate (DQ.CLAIMS_QUALITY_GATE)

A Dynamic Table (TARGET_LAG=DOWNSTREAM) that evaluates per-claim SNIP checks inline. Each claim gets a DQ_PASS or DQ_FAIL status. Gold edit rules only process DQ_PASS claims. DQ_FAIL claims receive a DQ_REJECT disposition in the final adjudication.

Per-claim checks in the DQ gate:

SNIP 1-2: NULL claim_id, Invalid NPI format, Invalid MBI format, Invalid TOB format, Future-dated admission
SNIP 3-4: Discharge before admission, Charges out of range, Inpatient missing admission source
SNIP 5-6: Invalid ICD-10 format

DQ Dashboard Views (5 views)

View	Purpose	Data Source
DQ.V_DQ_SCORECARD	Per-table, per-metric pass/fail with SNIP category and DQ dimension	SNOWFLAKE.LOCAL.DATA_QUALITY_MONITORING_RESULTS
DQ.V_DQ_TREND	Time-series of metric values for trend analysis	SNOWFLAKE.LOCAL.DATA_QUALITY_MONITORING_RESULTS
DQ.V_DQ_GATE_SUMMARY	Aggregate DQ_PASS/DQ_FAIL counts, percentages, SNIP breakdown	DQ.CLAIMS_QUALITY_GATE
DQ.V_DQ_QUARANTINE	DQ-failed claims with per-check flags for investigation	DQ.CLAIMS_QUALITY_GATE
DQ.V_DQ_DIMENSION_HEATMAP	Pass rates by SNIP category and DQ dimension per table	DQ.V_DQ_SCORECARD

DQ Dimensions tracked: COMPLETENESS, VALIDITY, CONSISTENCY, ACCURACY, INTEGRITY, TIMELINESS, VOLUME

7. Task 6: Gold Edit Rules Engine (11 Rules)

All edit rules are implemented as SQL-based Dynamic Tables with TARGET_LAG=DOWNSTREAM. Each rule joins DQ.CLAIMS_QUALITY_GATE filtering on dq_status=DQ_PASS. Results are aggregated into EDIT_RESULTS_SUMMARY and flow to CLAIM_ADJUDICATION (terminal DT).

Original 8 Pre-Payment Edit Rules

Code	Edit Name	Category	Severity	Logic
PPE-001	Duplicate Claim Check	DUPLICATE	REJECT	Same NPI+MBI+date+charges within 30 days
PPE-002	Provider Enrollment	PROVIDER	DENY	PECOS status must be Active; effective date <= submission
PPE-003	Beneficiary Eligibility	ELIGIBILITY	DENY	Part A status Active on admission date
PPE-004	Revenue Code Validation	CODING	REJECT	Revenue codes must exist in valid reference
PPE-005	Date Logic	DATE	REJECT	Discharge >= admission; admission <= submission
PPE-006	DRG Consistency	CODING	FLAG	DRG valid for TOB and principal diagnosis
PPE-007	NPI Validation	BILLING	REJECT	NPI exactly 10 digits, passes Luhn check
PPE-008	Timely Filing	DATE	DENY	Submission within 365 days of discharge

NEW: 3 Payer-Specific Medicare Edit Rules

Code	Edit Name	Category	Severity	Reference Table	Logic
PPE-009	MUE Check	PAYER	DENY	CONFIG.MUE_LIMITS	Units per day > max threshold for HCPCS
PPE-010	NCCI Unbundling	PAYER	DENY	CONFIG.NCCI_CODE_PAIRS	Billed code pairs that should be bundled
PPE-011	Frequency Check	PAYER	DENY	CONFIG.FREQUENCY_LIMITS	Service billed > max per rolling period

Adjudication Dispositions

Disposition	Meaning	Expected %
ACCEPT	Claim passes all DQ checks and all edit rules	~80%
DENY	Claim fails a DENY-severity edit (enrollment, eligibility, MUE, NCCI, freq, timely filing)	~10%
REJECT	Claim fails a REJECT-severity edit (duplicate, coding, format)	~5%
FLAG	Claim triggers a FLAG-severity edit (DRG consistency) - review recommended	<1%
DQ_REJECT	Claim failed SNIP 1-6 DQ gate - never entered edit rules	~5%

8. Technology Mapping

Requirement	Snowflake Feature	NRT Impact
S3 -> Bronze Ingestion	Snowpipe (auto_ingest + SQS)	~1-2 min latency
Bronze -> Silver Transform	Dynamic Tables (DOWNSTREAM)	Cascading, on-demand
SNIP 1-6 DQ Checks	27 DMFs (14 built-in + 13 custom)	Auto-evaluates on data change
DQ Gate (pre-edit filter)	Dynamic Table (CLAIMS_QUALITY_GATE)	SNIP-aligned per-claim pass/fail
DQ Dashboard	SNOWFLAKE.LOCAL.DATA_QUALITY_MONITORING_RESULTS + 5 views	Real-time DQ by SNIP category
Silver -> Gold Edit Rules	11 Dynamic Tables (DOWNSTREAM)	Pure SQL, DQ-gated
Payer-Specific Edits	Reference tables (CONFIG) + Gold DTs	Lookup-driven, configurable
Final Adjudication	Dynamic Table (5 MIN lag)	Terminal DT drives pipeline
Orchestration	Fully declarative (no Tasks in hot path)	Zero orchestration overhead
Data Architecture	Medallion (BRONZE/SILVER/GOLD)	Schema separation
RBAC	Custom roles + future grants	Least-privilege access

9. Implementation Tasks

#	Task	Scope
Task 1	Snowflake Infrastructure Setup	Database, schemas, RBAC, warehouses, external stage
Task 2	Synthetic Data Generation	~50K claims, ~200K lines, reference tables, 15% error seeding
Task 3	Bronze Layer (Snowpipe)	Auto-ingest pipes, raw landing tables, catch-up task
Task 4	Silver Dynamic Tables	DOWNSTREAM-lag DTs with new SNIP fields (adm_source, sex)
Task 5	SNIP-Aligned DMFs (27)	14 built-in + 13 custom, DQ quality gate DT
Task 6	DQ Dashboard Views	5 views powered by DATA_QUALITY_MONITORING_RESULTS
Task 7	Gold Edit Rules (11)	8 original + 3 payer-specific (MUE/NCCI/freq), adjudication DT
Task 8	Gold Analytics Views	Edit hit rates, disposition summary, provider risk, payer breakdown
Task 9	End-to-End Testing	Trickle-feed simulation, <15 min SLA validation, DQ + edit verification